A Pretrained Sequence Model of Company Events for Credit Risk

Jens Nylander and Marcus Thorelius

The Intelligence Company AB (publ)
jens@tic.io marcus@tic.io

October 2026

Abstract

A company credit score is usually a scorecard: a logistic regression over characteristics of the company, often with financial ratios from its annual accounts. For a company that files on time, the latest annual report can describe a year that ended up to nineteen months earlier, and many events happen after it: late VAT and payroll returns, penalties, tax registrations revoked for misconduct, changes in payroll. We train one model on every limited company in Sweden. Each company is a sequence of dated events from public registers, together with its relations to other companies and the national economy. The model reads no data about natural persons. It gives the probability of bankruptcy within 3, 12 or 24 months, of enforcement debt, of a tax registration revoked for misconduct and of an overdue annual report. We train it first on synthetic companies, then on the real register, then briefly on the questions, and adjust its probabilities to the observed rates of companies we used for comparison. On the test companies, a company that goes bankrupt within 12 months gets a higher score than one that does not 92.4 % of the time (an AUC of 0.924). For the TabICL baseline, a pretrained model for tables given 605 columns built from the same records, it is 91.3 %. The riskiest 10 % of companies by the model contain 76 % of the companies that go bankrupt within 12 months. The model has a higher AUC than the TabICL baseline on bankruptcy within 12 and 24 months, enforcement debt and an overdue annual report, and a higher PR-AUC, which weighs the highest-ranked companies most, on bankruptcy within 24 months. For each request the service reads the company's events known at that moment; its probabilities differ from the training code's by 0.0006 on average.

1 Introduction

Credit scores are typically built as scorecards [12, 15]. The developer groups each candidate variable into ranges, keeps the variables that predict best and fits a logistic regression; its output, the probability of a bad outcome within a fixed period such as 12 months, is turned into a score. For companies, the variables include financial ratios from the annual accounts [2]. A Swedish limited company must file its annual report within seven months of the end of its fiscal year,[1] and that report stays the latest one until the next is filed. For a company that files on time, the latest report can therefore describe a year that ended up to nineteen months earlier.

Many events happen between two annual reports: late VAT and payroll returns, a falling payroll, penalties, a revoked alcohol permit, a tax registration revoked for misconduct. A company can also share its management with a company known for misconduct. Altman et al. [2] showed in 2010, on 5.8 million sets of accounts of UK companies, that adding non-financial information (legal actions by creditors, filing history, audit opinions, company age) to the accounts raises the AUC on companies not used to fit the model from 0.67–0.71 to 0.75–0.76.

[1] The annual general meeting is held within six months of the fiscal year end (Companies Act (2005:551), ch. 7, § 10), and the certified annual report must reach Bolagsverket within one month of the meeting's adoption of the accounts (Annual Accounts Act (1995:1554), ch. 8, § 3).

In Sweden these events are recorded every day by public bodies: Bolagsverket (the Swedish Companies Registration Office), Skatteverket (the Swedish Tax Agency), Kronofogden (the Swedish Enforcement Authority), Länsstyrelsen (the County Administrative Board), which imposes sanction fees, Kammarkollegiet (the Legal, Financial and Administrative Services Agency), which collects sanction fees decided by other authorities, Folkhälsomyndigheten (the Public Health Agency of Sweden), which keeps the national register of the alcohol and tobacco permits that the municipalities issue, Arbetsförmedlingen (the Swedish Public Employment Service), which pays employment subsidies, Försäkringskassan (the Swedish Social Insurance Agency), which pays dental care compensation, and others. The authors work at The Intelligence Company AB, a credit reporting company with a permit from Integritetsskyddsmyndigheten (the Swedish Authority for Privacy Protection), which receives these records, so we can train a model on them.

Table 1 lists the main sources, examples of their company events, and what the model uses today. Each body holds more than the table lists: Bolagsverket's record layout for the data it delivers to credit reporting companies is 182 pages long.[2]

Companies are attractive tools for organised crime, and the trade in shelf companies and in companies bought for their history is a central part of its infrastructure [6]. Table 2 summarises the study of the Swedish National Council for Crime Prevention. One in five people in criminal networks was involved in a company, and those companies went bankrupt six times as often as companies in general [6]. The study describes two kinds of straw director: fronts who pass checks, and directors put in just before a bankruptcy to take the consequences. It also describes a trade in companies bought for their history, registrations and bank accounts. The government's strategy against organised crime puts incorrect welfare payments at SEK 15 to 20 billion in 2021 and the criminal economy at SEK 100 to 150 billion a year, and names businesses run by criminals inside the welfare system as one route [5]. Most of the signals in that description are events in our data: a debt to the state when it arises, a revoked tax registration and its cause, a board and owners replaced shortly before a bankruptcy. The model sees that a board changed, not who joined it; records about persons are handled by rules applied after the model.

We therefore treat a company's records as a history of events, in date order. Each event is a token, the unit the model reads, and each company's history is one training example. We first train the model on all histories without any question (pretraining) and then briefly on the questions (fine-tuning); a new question can also be answered from labelled example companies, without fine-tuning. Related companies (an owner company, a company owned through another) are tokens in the same history, marked with the relation. A board member or an owner who is a person is only an event of the company, without the person. The economy is one token. For each of the next 24 months the model gives the probability of each outcome, so any period up to 24 months can be read from it, and a calibrated probability per question.

This paper describes: (1) an event format with two dates per record, the event date and the known date, and a leak check that verifies on every training example that every input was known by the scoring date; (2) training that starts on synthetic companies from a simulator; (3) an evaluation with two test sets, test companies never used in training and an out-of-time test set a year after the last training outcome, against a base-rate baseline and the TabICL baseline on the same rows; (4) results for Sweden on six questions: bankruptcy within 24 months passes our rule, and we give the reasons the other five do not; (5) a Rust service and API running the GPU code that reads a company's events at request time and must give the same predictions as the training code. We have not found prior work that trains one model on the event histories of a country's full company register and checks the known date of every event (Section 2).

[2] `https://bolagsverket.se/download/18.46f4138717c599ee403d0383/1719815644584/postbeskrivning.pdf`

Table 1: Public sources of a company's events in Sweden, and what the model reads from each.

public body	company events	used in the model
Bolagsverket (the Swedish Companies Registration Office)	registrations and status; annual reports, late-filing fees and deficiency notices; registry cases; board changes and non-conformance; liquidation orders, liquidations, reconstructions, bankruptcies, striking-off; dividends; changes of name, purpose and articles; beneficial owners, groups, addresses	yes: 29 event kinds, 10 relation kinds
Kronofogden (the Swedish Enforcement Authority)	debts to the state and to others, enforcement cases and their outcomes, payment orders	yes: 15 event kinds
Skatteverket (the Swedish Tax Agency)	tax registration (income tax, VAT, employer) and its revocation with cause, late fees, surcharges, deferrals, monthly payroll declarations	yes: 7 event kinds
Statistics Sweden (SCB)	industry code	yes: 1 event kind
Folkhälsomyndigheten (the Public Health Agency of Sweden)	alcohol and tobacco permits	yes: 2 event kinds (serving permits)
Länsstyrelsen (the County Administrative Board)	penalties	no
Kammarkollegiet (the Legal, Financial and Administrative Services Agency)	penalties	no
Arbetsförmedlingen (the Swedish Public Employment Service)	employment subsidies	no
Försäkringskassan (the Swedish Social Insurance Agency)	dental care compensation	no
Lantmäteriet (the Swedish Mapping, Cadastral and Land Registration Authority)	property ownership	no
Transportstyrelsen (the Swedish Transport Agency)	vehicle fleets	no

Table 2: Companies tied to criminal networks in Sweden, 2020 to 2023, from the study of the Swedish National Council for Crime Prevention [6].

companies tied to criminal networks	figure
people in criminal networks involved in a company	about 11 000, one in five of the networks' people
companies they were involved in	just over 23 000, 1.7 % of all companies
share that were limited companies	about 80 %, against 65 % of all companies
bankruptcy rate	six times the general rate
overdue debts to the state	about SEK 11.5 billion
most common industries	construction, restaurants, real estate, trade
core group	fewer than 300 people, over 12 % of the companies, 19 involvements each on average
core group's companies bankrupt by 2025	one in four, against 3 % of all companies
criminals acting as auditors	about 30, for over 3 000 companies

2 Related work

Bankruptcy prediction from accounts. Altman et al. [2], on non-listed UK companies, is the reference result for what non-financial information adds to the accounts (see the introduction). The work closest to ours in data type is Aliaj et al. [1], who predict the bank default of Italian companies from the central credit register's quarterly loan records: about 800 000 companies over 2009 to 2014, with all reported results on the 300 000 companies with balance sheets and features from the last quarter of 2013 and the four quarters of 2014. Combinations of decision trees perform best. Their data describe a company's loans from banks; ours are public records about the company. They reduce each company to one row; we keep its history.

Scorecards in practice. Siddiqi [12] describes how scorecards are built and run in practice: grouping each variable into ranges, choosing the variables, fitting a logistic regression, turning it into points, testing and monitoring, with separate scorecards for different groups of companies where needed. Thomas et al. [15] cover the statistical methods, including methods for the time until default, and the rules for scoring that banks must follow under the Basel accords.

Pretrained models for tables. Some pretrained models for tables predict from labelled example rows given together with the question, without being trained on the task: TabPFN [7] and TabICL [9]. We use TabICL as a baseline, on 605 columns built from the same records as of the scoring date. A benchmark of databases with linked tables [11] shows that models which read across the linked tables do better than hand-built summary columns. Our tokens for related companies apply the same idea to event histories.

Network design. The recurrent layers we use [18] cost time in proportion to the length of the history. Qwen3-Next [10] uses three of them for each full attention layer, the layout we use. We let attention see the time between events with the method of Su et al. [13], using days instead of positions. To represent amounts from a few kronor to billions, we use the method of Tancik et al. [14] on a logarithmic scale. Calibration follows Zadrozny and Elkan [19], and the ranges around the probabilities follow Angelopoulos and Bates [3] and Vovk et al. [17].

3 Problem definition

Population and scoring date. The unit is the life of a limited company in Sweden: 1 473 494 companies in the export of 2026-10-15. A question is asked at a scoring date T. The model may

only read what was known at T. Each record has an event date and a known date, and only the known date decides whether the record is read.

The questions. Six questions are trained and tested. Each gives a probability for one company at T over a horizon h. The main question is *bankruptcy initiated within 12 months*. It is also asked at 24 months, and at 3 months on the out-of-time test set. Compulsory liquidation, company reconstruction and striking-off are separate outcomes: a company with one of them is counted under that outcome, not as a company that survived. A company whose horizon runs past the end of our data counts only for the months we can see. Three early-warning questions cover administrative events that often come before a bankruptcy. *Enforcement debt*: a new debt at Kronofogden, or an increase of an existing one, within h. *Tax registration revoked for misconduct*: Skatteverket ends the company's tax registration for misconduct, for example an income tax return not filed, deficient reporting or payment, a trading prohibition, abuse, neglect, no reply to an injunction, or household-service tax credits not repaid. A revocation at the company's own request or on administrative grounds does not count. A revocation caused by the bankruptcy itself is removed, because it would give away the answer. *Annual report overdue*: Bolagsverket's late-filing fee, which is how an overdue report is registered. The share of positives in the test rows is 0.91 % for bankruptcy within 12 months, 1.85 % within 24 months, 0.22 % within 3 months (out-of-time test set), 5.22 % for enforcement debt, 0.65 % for a misconduct revocation and 3.83 % for an overdue report.

Link to bankruptcy. We measured each early-warning event on the 779 155 limited companies active in Sweden on 2025-09-01, one event at a time, as the 12-month bankruptcy rate of the companies with the event divided by the rate of all of them, 1.24 %.[3] A first late-filing fee raises the rate 8.5 times, a second 9.4 times and a third 29.4 times. A tax registration revoked for misconduct raises it 26.4 times, and 32.8 % of those companies go bankrupt within the year. An overdue annual report raises it 17.7 times, and an open liquidation order 61.9 times.

One definition per label. Each question is defined once, in the evaluation code. The training code derives its labels from the same records, and the leak check verifies on every row that both agree. No rule changes a probability after the model and its calibration have produced it.

4 Data

Appendix A lists every input the model reads, generated from the source files.

Two dates per event. Only the known date decides whether an event is read (Section 3). The two dates can differ: a voluntary liquidation, for example, is registered some time after the meeting that decides it, and a model that read decision dates would use information that was not yet available. The leak check runs over every training example and verifies for every token that it was known by the scoring date, that an ended relation was known to have ended, that every related company was linked through a relation that was known and still active on that date, that the outcomes lie after the scoring date, and that the labels equal those of the evaluation code. In training and testing the scoring date lies in the past, so later records exist and must be left out; in use it is the day of the request. Sources that started late come with

[3] Computed from the training export. An event counts by its known date, in the nine months before 2025-09-01, which lie inside the period in which Bolagsverket's change notifications are complete. A late-filing fee counts by its number and only if it was not voided. A misconduct revocation uses the reason codes of the question. An annual report is overdue when the company has no report for the fiscal year that ended 7 to 19 months earlier, taking that year as the latest known report's period end plus twelve months. A liquidation order is open when it was neither recalled nor followed by compulsory liquidation.

records of when each source started, so "no debt" before a source started means "not collected", not "no debt".

Late sources. Several sources first reached us years after the events they describe, and their older records exist only for the companies that were still alive when the source was delivered. A model that read those records would learn that an old record means survival. The delivery check runs on the export. For every record kind and every year, it compares the share of companies with the record among those that went bankrupt within a year and among those still alive a year later. In the first export the check failed for the board history, the officer roles, the figures from data vendors, the official industry code, the late-filing fees and the addresses. From the second export on, every record counts from its source's first delivery date and a vendor's figures from the day they arrived, and the addresses and the group structure are left out. The export of 2026-10-15, on which the model is trained, passes the check on all 57 record kinds, five of them with a warning.

Event kinds. 54 event kinds (Appendix A). From Bolagsverket: annual reports received, approved and registered, the figures they contain, deficiency notices, findings in the reports, late-filing fees, dividend decisions, liquidation orders issued and recalled, registry cases received and closed, board gaps and composition changes, changes of name and purpose, and the outcome events: bankruptcy initiated, terminated and repealed, liquidation decided and ended, reconstruction decided and ended, striking-off. Officer roles started and ended, with the role code, and beneficial owners added and removed, with the extent and nature of control, are events of the company without the person. From Kronofogden: debts registered, changed and cleared, a debt already present at the authority's first delivery with an unknown start, enforcement cases registered, under execution, closed, withdrawn and cancelled, no-assets reports, and payment orders applied for and decided, with their later changes and removals, each case once. From Skatteverket: tax registration changes with the cause of a revocation, late fees on returns, surcharges, deferrals, and monthly payroll declarations and their corrections. From Statistics Sweden: the industry code in the 2025 version of the Swedish industry standard (SNI 2025), known from the day it reached us; the code for no registered activity gives no event. Every company also has an industry code inferred from its name and purpose text, with the code's probability, known from its registration; once a company has a registered code, the profile uses that code instead of the inferred one. The classifier is a text model [20] trained further on the companies with a registered SNI 2025 code; in training, each company's inferred code comes from a classifier that never saw the company. From Folkhälsomyndigheten: serving permits granted and ended. Each kind has its own fields: amounts, counts, days, and codes from fixed code lists with 2 849 values in all, each with a text description. The export of 2026-10-15 holds 69.3 million events for 1 473 494 limited companies, the inferred industry codes included. The largest kinds are payroll declarations (27.0 million), officer roles started (10.0 million) and the figures of filed reports (5.1 million).

Relations. 10 relation kinds and 0.90 million relations, all between companies. Most are beneficial ownership through an intermediate company (0.57 million) and auditor roles held by audit firms (0.24 million); the rest are other roles held by companies. Each relation has a start date, an end date and its details: the role code, and the extent and nature of control. For each company the model always reads its owner companies and audit firms, and at most 24 other related companies, of which up to eight are companies owned through it. A role held by a person, or an owner who is a person, is not a relation but an event of the company (above). The addresses and the group structure were left out after the delivery check: the addresses kept their sources' own dates, decades back, and the group structure was a single snapshot. This also removed the related companies they gave: companies at the same address, parents, subsidiaries

and sister companies. Links between co-debtors and creditors are not exported and not in the model.

Accounts. A fixed set of 55 figures per filed statement, merged from vendor, digital and scanned sources, each with a value and a unit. A vendor's figures are known from the day they arrived, and a report without a registration date is known twelve months after its period end. The report record also has the fiscal period and its length, the accounting framework, interim and consolidated flags, the source and precision of the figures, the auditor's remark, and the change in net sales and operating result from the previous year. Findings in a report are separate events with a kind, a weight, an amount and a date. A filing-state record gives the latest filing's age, the months to the next deadline, the number of filings known, the number of loss years in a row, ratios (equity ratio adjusted for untaxed reserves; cash, short-term liabilities and bank debt to total assets; interest coverage) and flags (loss, negative equity, equity below half the share capital, report overdue and for how long). An industry comparison record gives, for eight key figures from Statistics Sweden, the industry's quartiles for the company's size band, the company's own value, its difference from the median and its position.

State and economy. A profile record at the scoring date: legal form, age, industry code and section, size band by employees, municipality of the registered office, and the number of known events and relations. Active periods, with the days in each state: board gap, enforcement debt outstanding with amount and cases, tax registration withdrawn, VAT and employer registration, tax deferral outstanding, liquidation, reconstruction. A calendar record: the month, the days to the next filing deadline, the days since the fiscal year end. For the economy, 26 national series as published at the scoring date: unemployment on two definitions, GDP on three, import prices, consumer prices on four variants, house prices nationally and in eleven regions, the policy rate, the three-month treasury bill, the ten-year government bond and the five-year mortgage bond. Each series enters with its latest value and its twelve-month change, read by the time-series model Chronos-2 [4]. The production index of the company's own industry is a separate record.

What we do not read. Nothing about natural persons. The export has no person identifier: a role or an owner is only an event of the company, so board size and board turnover are company facts, but the model cannot tell whether two companies share a director. A person's own records, and what happened to the person's other companies, are handled by rules applied after the model. Also left out: the addresses, the group structure and the articles of association (the articles table has no history of changes). Available but not yet used: the property and vehicle registers, the text of annual reports, and the full set of tagged facts in digital reports.

5 Model

Events, fields, tokens. An event has fields of fixed types: a code, an amount, a count, a date, a period. Amounts and counts are turned into vectors on a logarithmic scale [14], so that small and large amounts are both represented. Every code starts as the vector that a text model, Qwen3-Embedding-0.6B [20], gives its description, and training adds a learned correction. A small attention layer combines the fields of an event into one token.

Network. The network has 18 blocks, and each token is a vector of 768 numbers. Three of every four blocks are recurrent layers [18], whose cost grows in proportion to the number of events; the fourth is a full attention layer, which can look at any token directly. Each token carries its date in days before the scoring date [13], so the network sees the time between events. Half of the attention looks at all tokens, of the company and of its related companies; a quarter only at tokens of the same company; and a quarter at the company itself and at tokens of the

same registry case. A token for a related company has the kind of relation, its direction and how many steps away the company is. The network has 232 million parameters.

The economy. A pretrained time-series model, Chronos-2 [4], used as it is, turns the 26 series as published at the scoring date into one token next to the company's events. We have not measured how much the model uses this token. Industry-level series are the next step.

Outputs. One output gives, for each of the next 24 months, the probability that the company goes bankrupt, goes into compulsory liquidation, enters reconstruction or is struck off in that month, if none of them has happened before [16]. A second output gives the same month-by-month probabilities for enforcement debt, revoked tax registration and overdue annual report. The question itself is given to the model as tokens: the company, the event asked about, the horizon and the scoring date.

What pretraining predicts. The next event and when it happens; which events happen in the coming year, as up to 16 guesses matched to the real events in the best order [8]; fields hidden from the input; the figures of the next annual report; and the month-by-month probabilities of the outcomes. Two further training terms shape how the network groups similar companies. The inputs are what was known at the scoring date, and everything predicted lies after it.

6 Training

Synthetic companies first. A simulator generates populations of companies: a hidden health for each company, economic cycles, group structures and shared addresses along which distress spreads, filing calendars with fees, consistent accounts, the four outcomes, and sources that publish late. The simulator uses overall rates of the real market, never individual companies. The model trains on 320 such populations, 12 500 steps and 6.5 billion tokens, before it sees a real company. The aim is that it first learns general patterns: warning events come before bankruptcies, distress spreads along relations, and a recession raises every rate.

Pretraining on Swedish companies. All companies except the test and validation companies: 10 625 steps from where the synthetic stage ended and 5.5 billion tokens, of which 4.9 billion from Swedish companies and the rest from the synthetic populations, which stay in the mix. We train on eight B200 GPUs.

Fine-tuning. One fine-tune for the five questions on the test companies and one for the out-of-time question, in 478 and 604 seconds on eight GPUs. One training company in ten is a validation company, kept aside to choose and calibrate. We stop at the step where the calibrated probabilities fit the validation companies best: step 200 for the questions on the test companies and step 500 for the out-of-time question. The out-of-time question has its own split in time, with a gap between the training and test months.

Calibration and ranges. The model's outputs are turned into probabilities by a step function that only goes up, fitted on the validation companies so that each step's probability is the observed rate of its companies, with at least 50 validation rows per step. The probabilities are then shifted to match one of two rates: the rate of the most recent months, or the long-run rate since 1990 for companies of the same age and industry. Each probability also gets a 90 % range, computed from the errors on the validation companies.

Answers from examples. A small extra layer can answer a question without fine-tuning: it is given labelled example companies together with the question and predicts from them. It is trained on such tasks from the synthetic populations and from part of the pretraining data.

Industry codes of 2025, one candidate. The model is our third training run. Statistics Sweden now assigns industry codes in SNI 2025 only, so our second run, which read the codes of the 2007 version, had no registered code for the newest companies. The third run reads SNI 2025 codes only, the code inferred from the purpose text where a company has none, and its classifier is trained on SNI 2025 codes. The rest follows the second run. There is one pretraining run, and the fine-tuning steps were chosen on the validation companies before any test row was read. The second run's reported model was chosen between two candidates after its test rows had been read; Section 8 compares it with the model on the same test rows.

7 Evaluation

Every model is scored on the same rows: the same companies, scoring dates, outcomes and weights, by evaluation code that takes any file of predictions. There are two test sets. The test companies are one company in ten, chosen by a fixed rule from the company's number, never used in any training stage, and scored at twelve monthly scoring dates. The out-of-time test set has test months a full year after the last training outcome, with no overlap in outcomes. Two baselines are scored on the same rows. The base-rate baseline gives every company the training rate of its age and industry group, pulled towards the overall rate for small groups; a model whose probabilities are further from the outcomes than this baseline's is not calibrated. The TabICL baseline builds 605 columns from the same records as of the scoring date and scores them with TabICL [9], given a sample of training rows with as many events as non-events; its probabilities are then corrected back to the real base rate. We report:

- AUC: the probability that a company with the event gets a higher score than a company without it; 0.5 is chance and 1 is perfect;
- PR-AUC: how many of the highest-scored companies really have the event, averaged over list lengths; it weighs the top of the ranking most;
- Brier skill: how much closer the probabilities are to the outcomes than the base rate is, where a positive value means closer;
- calibration error: the average gap between the predicted probability and the observed rate, over ten groups of companies sorted by probability;
- the share of the companies with the event that are among the riskiest 1, 5 and 10 % of companies.

All are weighted to the population. A question passes when the model has both a higher AUC and a higher PR-AUC than the TabICL baseline on the same rows.

8 Results

Table 3 shows the results. For bankruptcy within 12 months the model has an AUC of 0.924, the TabICL baseline 0.913 and the base-rate baseline 0.717. The riskiest 10 % of companies by the model contain 76 % of the companies that go bankrupt within 12 months; for the TabICL baseline the share is 73 % and for the base-rate baseline 33 % (Table 4, Figure 1).

The model has a higher AUC than the TabICL baseline on bankruptcy within 12 months (0.924 vs. 0.913), bankruptcy within 24 months (0.890 vs. 0.853), enforcement debt (0.891 vs.

0.884) and annual report overdue (0.8944 vs. 0.8936). It has a lower AUC on tax registration revoked (0.951 vs. 0.952) and on bankruptcy within 3 months on the out-of-time test set (0.924 vs. 0.927). It has a higher PR-AUC on bankruptcy within 24 months (0.201 vs. 0.149), which passes. On the other five it has a lower PR-AUC: bankruptcy within 12 months (0.254 vs. 0.256), enforcement debt (0.508 vs. 0.520), tax registration revoked (0.195 vs. 0.243), annual report overdue (0.396 vs. 0.400) and bankruptcy within 3 months on the out-of-time test set (0.053 vs. 0.095).

Calibration lowers PR-AUC. Before calibration the model's PR-AUC is higher than the TabICL baseline's for bankruptcy within 12 months (0.268 vs. 0.256), enforcement debt (0.5204 vs. 0.5201) and annual report overdue (0.406 vs. 0.400). Calibration keeps the order of the companies, but all companies on the same step get the same probability. These ties lower PR-AUC more than AUC, because PR-AUC depends on the order among the riskiest companies. For bankruptcy within 12 months the AUC is 0.924 both before and after calibration. After calibration every question has a positive Brier skill and a calibration error of 0.0051 or less (Figure 2).

Table 3: Results. Bankruptcy within 3 months is tested on the out-of-time test set, the other questions on the test companies. Positives: the share of test rows with the event. Base-rate: the base-rate baseline.

		base-rate	TabICL		the model, calibrated			
question	positives	AUC	AUC	PR-AUC	AUC	PR-AUC	top 10 %	calib. error
Bankruptcy, 12 m	0.91 %	0.717	0.913	0.256	0.924	0.254	76 %	0.0009
Bankruptcy, 24 m	1.85 %	0.720	0.853	0.149	0.890	0.201	64 %	0.0007
Bankruptcy, 3 m, out of time	0.22 %	0.720	0.927	0.095	0.924	0.053	80 %	0.0005
Enforcement debt, 12 m	5.22 %	0.676	0.884	0.520	0.891	0.508	64 %	0.0051
Tax registration revoked, 12 m	0.65 %	0.734	0.952	0.243	0.951	0.195	87 %	0.0017
Annual report overdue, 12 m	3.83 %	0.626	0.894	0.400	0.894	0.396	64 %	0.0030

Table 4: Share of the companies that go bankrupt within 12 months that are among the riskiest 1, 5 and 10 % of companies, test companies.

companies	base-rate baseline	TabICL baseline	the model
riskiest 1 %	4 %	34 %	35 %
riskiest 5 %	19 %	61 %	63 %
riskiest 10 %	33 %	73 %	76 %

Records from before a source's first delivery. Our first run read the older records of late sources (Section 4). Table 5 compares the AUC of the first and the second run on the same test rows, with those records (first export) and without them (second export). Both our network and the TabICL baseline have their largest drop on bankruptcy within 24 months. On the three early-warning questions our network drops more than the TabICL baseline. On the out-of-time test set our network's AUC goes up.

Industry codes of 2025. Table 6 compares the second run, which read the industry codes of SNI 2007, with the model, which reads SNI 2025 codes only, on the same test rows (Section 6). The model has the same AUC on bankruptcy within 12 months (0.924 vs. 0.924), enforcement debt (0.891 vs. 0.891) and annual report overdue (0.894 vs. 0.894). It has a lower AUC on bankruptcy within 24 months (0.890 vs. 0.894), tax registration revoked (0.951 vs. 0.955) and the out-of-time test set (0.924 vs. 0.937).

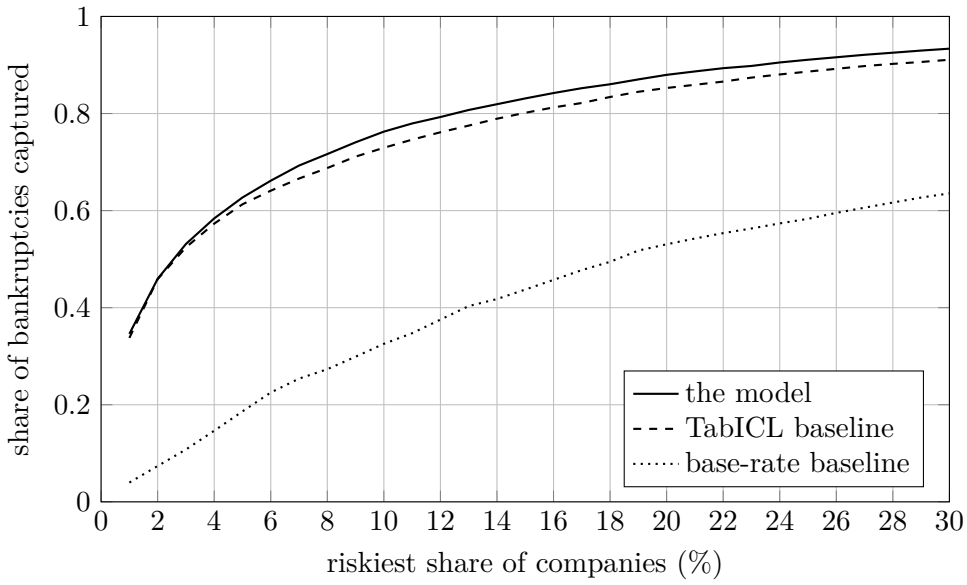


Figure 1: Share of the companies that go bankrupt within 12 months that are among the riskiest companies, by the share of companies taken, test companies.

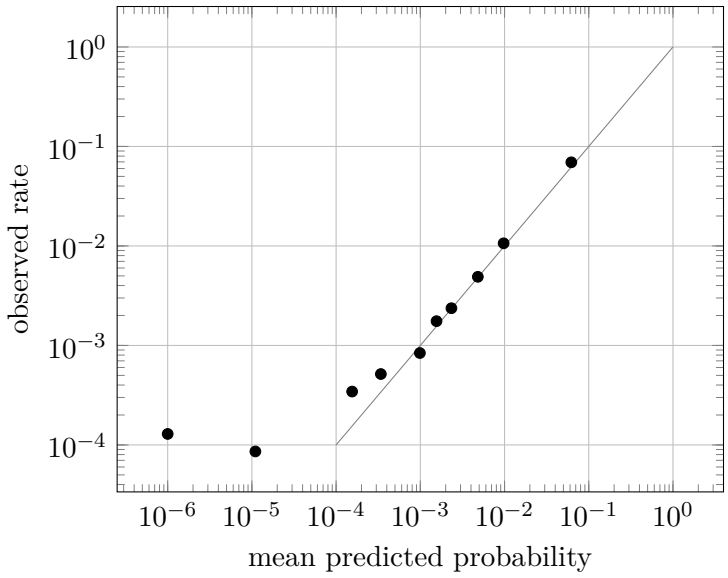


Figure 2: Observed rate against mean predicted probability, ten groups of companies sorted by probability, the model after calibration, bankruptcy within 12 months. The line is where the two are equal.

Table 5: AUC of the first and the second run on the same test rows, first export (with the older records of late sources) and second export (every record counted from its source's first delivery date). The enforcement-debt test set has 225 607 rows in the first export and 224 736 in the second.

question	our network		TabICL baseline	
	first	second	first	second
Bankruptcy, 12 m	0.927	0.924	0.915	0.910
Bankruptcy, 24 m	0.917	0.894	0.908	0.864
Bankruptcy, 3 m, out of time	0.923	0.937	0.930	0.931
Enforcement debt, 12 m	0.901	0.891	0.884	0.883
Tax registration revoked, 12 m	0.961	0.955	0.956	0.953
Annual report overdue, 12 m	0.905	0.894	0.895	0.891

Table 6: AUC and PR-AUC on the same test rows, the second run (industry codes of SNI 2007) and the model (SNI 2025 only), calibrated.

question	SNI 2007 (second run)		SNI 2025 (the model)	
	AUC	PR-AUC	AUC	PR-AUC
Bankruptcy, 12 m	0.924	0.246	0.924	0.254
Bankruptcy, 24 m	0.894	0.218	0.890	0.201
Bankruptcy, 3 m, out of time	0.937	0.061	0.924	0.053
Enforcement debt, 12 m	0.891	0.513	0.891	0.508
Tax registration revoked, 12 m	0.955	0.235	0.951	0.195
Annual report overdue, 12 m	0.894	0.395	0.894	0.396

Out-of-time test set. For bankruptcy within 3 months on the out-of-time test set, the model has a lower AUC than the TabICL baseline (0.924 vs. 0.927) and a lower PR-AUC (0.053 vs. 0.095). The riskiest 1 % of companies by the TabICL baseline contain 43 % of the bankruptcies; by the model, 32 %. All training months of this split lie before the newest event kinds started, while its validation and test months include them. We expect this to improve when we retrain with a full training period of those sources, without changing the model.

Answers from examples. Without fine-tuning, from labelled training rows given with the question, the answers have a lower AUC than the fine-tuned model on all three early-warning questions (Table 7).

Table 7: Answers from labelled examples against the fine-tuned model, early-warning questions, test companies.

question	fine-tuned		from examples	
	AUC	top 10 %	AUC	top 10 %
Enforcement debt, 12 m	0.891	64 %	0.881	62 %
Tax registration revoked, 12 m	0.951	87 %	0.949	86 %
Annual report overdue, 12 m	0.894	64 %	0.863	58 %

Errors found by the evaluation. We found five errors and fixed each. First, a fine-tune that ran too long kept its AUC but its probabilities no longer matched the observed rates; we now stop at the step where the calibrated probabilities fit the validation companies best, not at the best AUC. Second, two leaks between the questions' test sets: a joint fine-tune that included the out-of-time test set, and pretraining labels that ran to the end of the data. Both made the

test results too high. Both are fixed, and the build now fails if either comes back. Third, a scoring bug read the probability of any of the four outcomes for every question, including the early-warning questions, which have their own output. The small-scale result tables with this bug are kept and marked as replaced. Fourth, in the data: the delivery check found sources whose older records exist only for companies that survived, and the first run read them (above). Fifth, in the third run the leak check stopped the build: related companies whose only record had been their SNI 2007 code had no record left in the export but still took a place among the related companies of a sample. Such a company now gets no place, in training and in the service alike.

One company. Figure 3 shows the events of one anonymised test company and the model's calibrated probability of bankruptcy within 12 months, month by month over the three years up to its last test scoring date. The probabilities are the service's answer for the company today: each month is scored from the records known on that date. The company was about seven years old at the last scoring date. In the two years before that date the registers show two annual reports with the finding that the company was obliged to go into liquidation, one of them also with a late annual general meeting, a new board member and deputy, a change of name and purpose, registry cases, a new board member again, a change of owners, two late fees on tax returns, a surcharge on VAT, a payment order and a board gap. The probability stayed between 0.25 and 1.8 % until seven months before the last scoring date, in risk class 3 to 5. It was 7.4 % at six months, 14 % at five, 19 % at two, 30 % at one and 57 % at the last scoring date. The risk class was 1 from six months before. The model's output before calibration (dashed line) went from 0.10 three years before to 0.94. The enforcement debt, the report that the company had no assets, the tax registration revoked for misconduct and the late-filing fees for the annual report all became known after the last scoring date: the debt in month 0 of the figure, the no-assets report in month +1, the revocation and the first late-filing fee in month +2 and the second in month +4. Bankruptcy was initiated in month +4. The figure is drawn from the company's records and the service's scores for it. Events are placed in the month they became known, amounts are shown as bands, and no identifier is kept.

9 Serving

The service is written in Rust and runs the trained network on the GPU with GPU code we wrote ourselves, so before a model is served, the service must give the same predictions as the training code on the test rows. On all 192 645 test rows of the 12-month bankruptcy question, the service's AUC before calibration is 0.92447 and the training code's 0.92446. The mean difference in probability is 0.0006 and the largest 0.015. In the first run this check found one bug in our rewrite of the training code. A second check compares the service's calibrated probability and range with the training code's calibration on one company at ten scoring dates; it found that the ranges were applied to the probability before calibration instead of after. Both bugs are fixed and both checks are kept.

The service stores no company data. For each request it reads the events of the company and of its related companies, with the same code that wrote the training export, so a score uses every event known at that moment. It infers the industry code of every company it reads with the same text classifier, as the mean of its five versions, turns the events into tokens with the training vocabulary, and scores them.

10 Limitations and open items

Five questions do not pass. On bankruptcy within 12 months, enforcement debt, tax registration revoked, annual report overdue and the out-of-time test set, the model has a lower PR-AUC

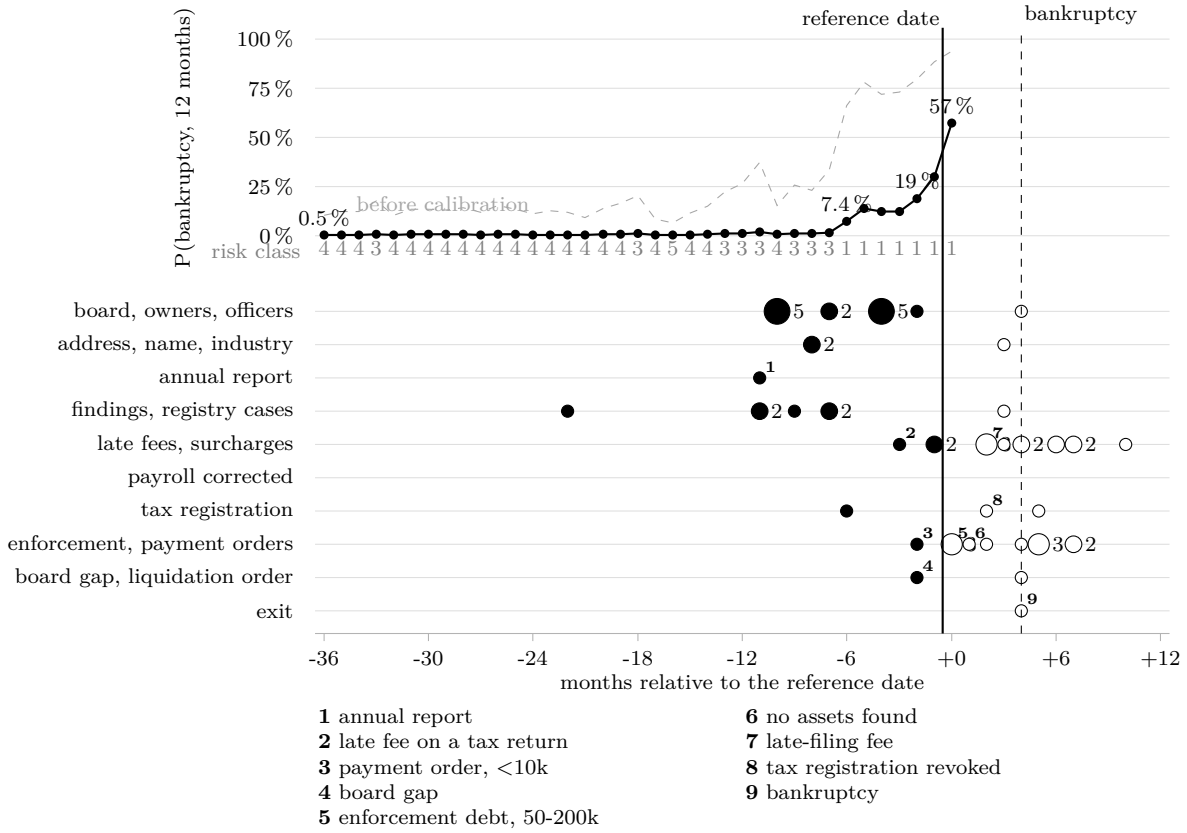


Figure 3: One anonymised test company, scored by the service month by month over three years. Top: the probability of bankruptcy within 12 months at each monthly scoring date, after calibration (solid) and before (dashed), with the risk class under each point. Bottom: the company's events per month by type; the dot size is the count, and the decisive events are numbered and named. Each probability uses only the events to its left. The vertical line is the company's last test scoring date, whose view the strip shows: filled dots were known by then, open dots came later. Months are relative to that date, each event is in the month it became known, and amounts are bands.

than the TabICL baseline (Section 8). The out-of-time question cannot yet train on the newest event kinds. A test company's related companies can be training companies; this only affects values computed over the related companies' own related companies, and we accept it for this test set. We have not measured how much the model uses the economy token. There are no bank default labels, so bankruptcy is used instead of default. There is no trade payment data, the records of how a company pays its suppliers. Data about persons is excluded by decision; from the second export on, the export does not even have the anonymous person relations of the first. Reasons, monitoring, behaviour tests and lender adapters are designed but not built.

Calibration is a step function. In the first run some steps at the top of the scale covered only a few validation rows, so the probability jumped between nearby model outputs. Each step now covers at least 50 validation rows, and rows with the same model output are never split between two steps. For the 12-month question the model has 156 steps on 49 218 validation rows. Its top step is 87 %: that share of the validation companies in its riskiest step went bankrupt within the year. The example in Figure 3 uses the refitted calibration.

11 Reproducibility

The evaluation code scores any file of predictions on the fixed rows. Every table, figure and number in this paper is generated from the evaluation output, the training export, the training logs and the service's own checks; apart from the cited figures, no number is typed by hand.

A Input inventory

The tables in this appendix are generated from the files that define what the model reads: the descriptions of the event kinds, the figure set, the limits on related companies, the code lists, the run's list of economy series and the run's export, so that only kinds the export holds are listed. The descriptions are the English texts the text model turns into each kind's starting vector. In all: 54 event kinds, 10 relation kinds, 55 figures per filed statement plus 8 industry comparison figures, 2,849 code values used by the events, and 26 national economy series.

A.1 Event kinds

kind	description given to the text model
Exits	
`bankruptcy.initiated`	bankruptcy initiated
`bankruptcy.terminated`	bankruptcy terminated
`bankruptcy.repealed`	bankruptcy repealed by the court
`liquidation.decided`	liquidation decided
`liquidation.ended`	liquidation ended
`reconstruction.decided`	company reconstruction decided
`reconstruction.ended`	company reconstruction ended
`status.struck_off`	company struck off the register
Registry: filings, fees, cases, board, changes	
`report.received`	annual report received by the registry
`report.approved`	annual report approved by the registry
`report.registered`	annual report registered
`report.shortage`	deficiency notice on the annual report
`fee.registered`	late-filing fee imposed
`dividend.registered`	dividend decision registered
`report.finding`	finding in the annual report
`report.figures`	annual report figures (income statement, balance sheet, key ratios)

kind	description given to the text model
`liquidation_order.issued`	liquidation order issued by the registry
`liquidation_order.recalled`	liquidation order recalled
`case.received`	registry case received
`case.terminated`	registry case closed without registration
`board.gap_started`	board or representative gap started
`board.gap_ended`	board or representative gap ended
`board.composition`	board composition changed
`name.changed`	company name changed
`purpose.changed`	business purpose changed
Enforcement: debts, cases, payment orders	
`debt.registered`	debt registered with the enforcement authority
`payment_order.changed`	payment order changed (contested, withdrawn, dismissed or to court)
`payment_order.removed`	payment order removed
`debt.present`	debt with the enforcement authority at the first snapshot, start unknown
`debt.changed`	debt with the enforcement authority changed
`debt.cleared`	debt with the enforcement authority cleared
`enforcement.case_registered`	enforcement case registered
`enforcement.case_under_enforcement`	enforcement case under execution
`enforcement.case_closed`	enforcement case closed
`enforcement.case_partly_withdrawn`	enforcement case partly withdrawn
`enforcement.case_cancelled`	enforcement case cancelled
`enforcement.no_assets_report`	enforcement report: no attachable assets
`enforcement.decision_cancelled`	enforcement decision cancelled
`payment_order.applied`	payment order applied for against the company
`payment_order.verdict`	payment order verdict
Tax: declarations, fees, registration	
`payroll.declared`	employer payroll declaration filed
`payroll.corrected`	employer payroll declaration corrected
`tax.late_fee_return`	late fee for a tax return
`tax.late_fee_income`	late fee for the income tax return
`tax.surcharge`	tax surcharge imposed
`tax.deferral`	temporary tax deferral
`tax.registration`	tax registration changed (income tax, VAT, employer)
Statistics and permits	
`industry.code`	industry code registered
`permit.serving_granted`	serving permit granted
`permit.serving_ended`	serving permit ended
Officers and owners, as company facts without the person	
`role.started`	an officer took up a role (board, CEO, auditor, signatory)
`role.ended`	an officer left a role
`ownership.owner_added`	a beneficial owner was added
`ownership.owner_removed`	a beneficial owner was removed

A.2 Relations and limits on related companies

relation kind	description
`role.board_member`	board member
`role.proprietor`	proprietor
`role.partner`	partner
`role.limited_partner`	limited partner
`role.liquidator`	liquidator
`role.auditor`	auditor
`role.deputy_auditor`	deputy auditor
`role.external_signatory`	external signatory
`role.other`	other officer
`ownership.beneficial_via`	beneficial owner through an intermediary company
Paths to related companies (the tag on a related company's tokens)	
`focal`	the company itself
`owner`	an owner company (beneficial ownership through a company)
`owned`	a company owned through the company
`auditor`	the audit firm
Limits on related companies per company	
owners, auditors	always included
other related companies	at most 24 in all
companies owned through the company	at most 8

A.3 State and context records

record	description
`state.profile`	the company's profile at the as-of time
`state.interval`	an active state at the as-of time
`state.filing`	the latest filing and the fiscal year
`state.sector`	comparison with industry key figures
`state.coverage`	source coverage at the as-of time
`context.economy`	economy series, latest published value
`context.industry`	industry production series, latest published value
`query`	the question put to the model
Closed codes	
`profile.legal_form`	AB, HB, KB, EF
`sector.position`	below_q1, within, above_q3
`coverage.state`	live, not_yet, ended, undated

A.4 Figures of a filed statement

Every figure enters with its value and unit; the report record also has the fiscal period, the accounting framework, the source and precision of the figures and the auditor's remark. The eight industry comparison figures enter with the industry quartiles for the company's size band and the company's position against them. The model also reads "change in net sales against the previous year" and "change in operating result against the previous year" as named fields.

figure	unit	figure	unit
net sales	currency	total assets	currency
other operating income	currency	share capital	currency
total operating income	currency	unrestricted equity	currency
raw materials and consumables	currency	total equity	currency
goods for resale	currency	untaxed reserves	currency
other external costs	currency	long-term bank debt	currency
personnel costs	currency	total long-term liabilities	currency
depreciation and amortisation	currency	short-term bank debt	currency
total operating expenses	currency	accounts payable	currency
operating result	currency	tax liabilities	currency
financial income	currency	total short-term liabilities	currency
financial expenses	currency	total equity and liabilities	currency
net financial items	currency	average number of employees	count
result after financial items	currency	floating charges	currency
appropriations	currency	property mortgages	currency
result before tax	currency	pledged securities	currency
tax on the result	currency	contingent liabilities	currency
net result	currency	overdraft facility granted	currency
intangible assets	currency	proposed dividend	currency
tangible assets	currency	operating margin	ratio
financial assets	currency	gross margin	ratio
total fixed assets	currency	net margin	ratio
inventories	currency	return on equity	ratio
accounts receivable	currency	return on assets	ratio
total current receivables	currency	equity ratio	ratio
current investments	currency	debt-to-equity ratio	ratio
cash and bank	currency	quick ratio	ratio
total current assets	currency		

A.5 Code lists

Each event's code comes from one of these lists; every code has a text description, which the text model turns into the code's starting vector.

code table	codes
registry case types	336
liquidation order causes	6
liquidation order recall causes	11
officer role codes	32
deficiency notice reasons	91
company status codes	48
serving permit categories	52
enforcement debt types	41
enforcement case termination causes	15
no-assets report reasons	3
annual report findings	48
municipalities	290
tax registration revocation causes	16
industry codes, SNI 2025 (national standard, all levels)	1,860

A.6 Economy series

Each series enters as published at the scoring date, month by month, through the time-series model.

group	series	count
Labour market	unemployment rate, two definitions (monthly)	2
Output	GDP indicator, seasonally adjusted (monthly)	1
Output	GDP volume change, two bases (quarterly)	2
Prices	import price index (monthly)	1
Prices	consumer price index, four variants (monthly)	4
Housing	house price index, national and eleven regions (quarterly)	12
Rates	policy rate (daily, averaged to months)	1
Rates	3-month treasury bill (daily)	1
Rates	10-year government bond (daily)	1
Rates	5-year mortgage bond (daily)	1
all series		26

Acknowledgements

References

[1] Tesi Aliaj, Aris Anagnostopoulos, and Stefano Piersanti. Firms default prediction with machine learning. *arXiv preprint arXiv:2002.11705*, 2020.

[2] Edward I. Altman, Gabriele Sabato, and Nicholas Wilson. The value of non-financial information in SME risk management. *Journal of Credit Risk*, 6(2):95–127, 2010. doi: 10.21314/JCR.2010.110.

[3] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.

[4] Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.

[5] Government Offices of Sweden. Resistance and decisive action: a national strategy against organised crime. Technical Report Government Communication 2023/24:67, Government Offices of Sweden, 2024.

[6] Li Hammar, Henrik Angerbrandt, Johan Dahlberg, Marianne Danielsson, and Linnea Schleyer. Företag som brottsverktyg. Technical Report Rapport 2025:20, Brottsförebyggande rådet (Brå), 2025. Published 11 December 2025.

[7] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.

[8] Harold W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955. doi: 10.1002/nav.3800020109.

[9] Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICL: A tabular foundation model for in-context learning on large data. *arXiv preprint arXiv:2502.05564*, 2025.

[10] Qwen Team. Qwen3-Next: Towards ultimate training & inference efficiency. Release post, September 2025; model card at `https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct`, 2025.

[11] Joshua Robinson, Rishabh Ranjan, Weihua Hu, Kexin Huang, Jiaqi Han, Alejandro Dobles, Matthias Fey, Jan E. Lenssen, Yiwen Yuan, Zecheng Zhang, Xinwei He, and Jure Leskovec. RelBench: A benchmark for deep learning on relational databases. *arXiv preprint arXiv:2407.20060*, 2024.

[12] Naeem Siddiqi. *Intelligent Credit Scoring: Building and Implementing Better Credit Risk Scorecards*. Wiley and SAS Business Series. Wiley, second edition, 2017. ISBN 978-1-119-27915-0.

[13] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021.

[14] Matthew Tancik, Pratul P. Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. *arXiv preprint arXiv:2006.10739*, 2020.

[15] Lyn Thomas, Jonathan Crook, and David Edelman. *Credit Scoring and Its Applications*. Mathematics in Industry. SIAM, Philadelphia, second edition, 2017. doi: 10.1137/1.9781611974560.

[16] Gerhard Tutz and Matthias Schmid. *Modeling Discrete Time-to-Event Data*. Springer Series in Statistics. Springer, 2016. doi: 10.1007/978-3-319-28158-2.

[17] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.

[18] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule. *arXiv preprint arXiv:2412.06464*, 2024.

[19] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.

[20] Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025.